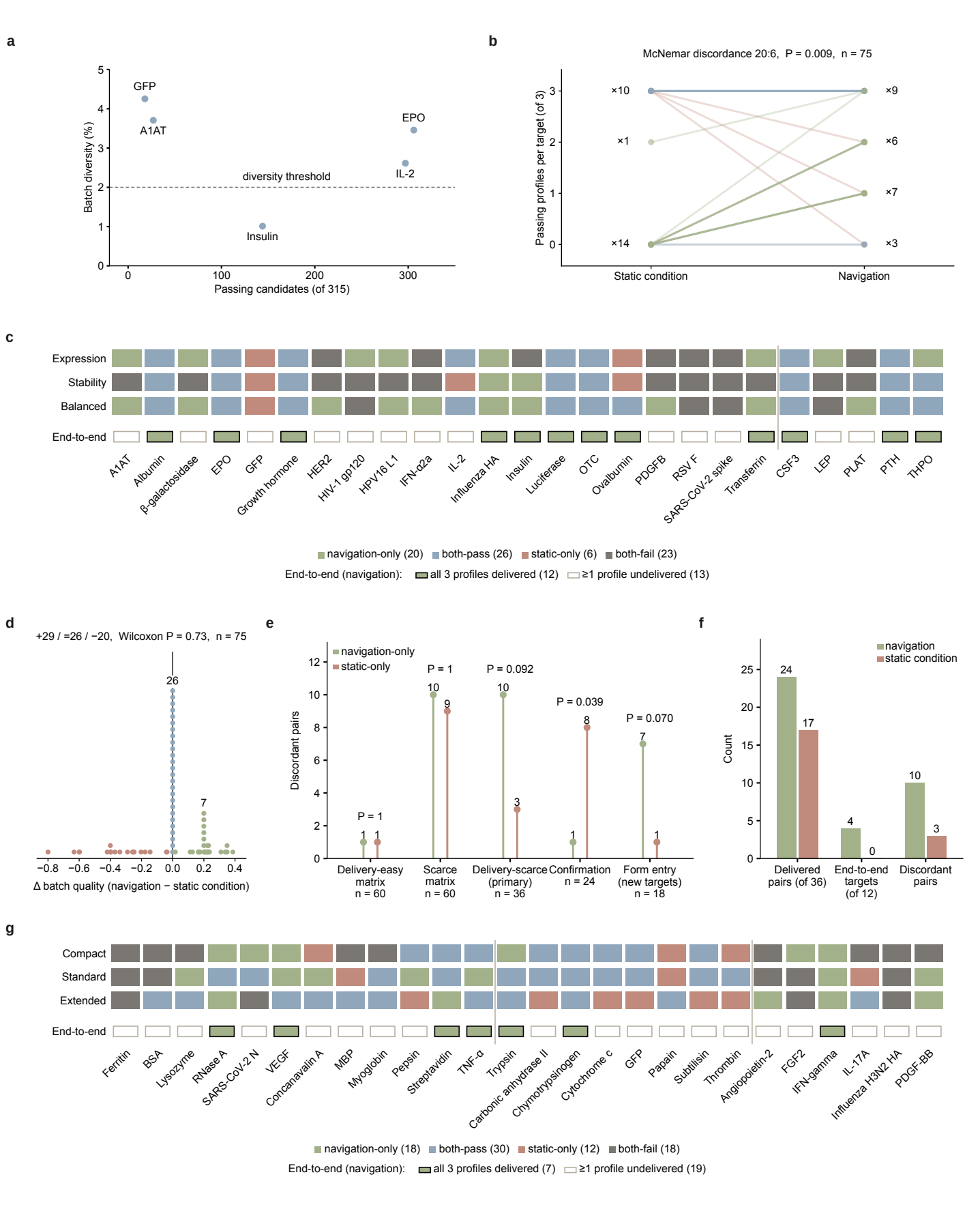

a
Batch diversity (%)
Passing candidates (of 315)
GFP
A1AT
EPO
IL-2
Insulin
diversity threshold
b
McNemar discordance 20:6, P = 0.009, n = 75
Passing profiles per target (of 3)
×10
×1
×14
×9
×6
×7
×3
Static condition
Navigation
c
Expression
Stability
Balanced
End-to-end
A1AT
Albumin
β-galactosidase
EPO
GFP
Growth hormone
HER2
HIV-1 gp120
HPV16 L1
IFN-α2a
IL-2
Influenza HA
Insulin
Luciferase
OTC
Ovalbumin
PDGFB
RSV F
SARS-CoV-2 spike
Transferrin
CSF3
LEP
PLAT
PTH
THPO
navigation-only (20)
both-pass (26)
static-only (6)
both-fail (23)
End-to-end (navigation):
all 3 profiles delivered (12)
≥1 profile undelivered (13)
d
+29 / =26 / −20, Wilcoxon P = 0.73, n = 75
Δ batch quality (navigation − static condition)
e
navigation-only
static-only
Discordant pairs
P = 1
P = 1
P = 0.092
P = 0.039
P = 0.070
Delivery-easy matrix n = 60
Scarce matrix n = 60
Delivery-scarce (primary) n = 36
Confirmation n = 24
Form entry (new targets) n = 18
f
navigation
static condition
Count
Delivered pairs (of 36)
End-to-end targets (of 12)
Discordant pairs
g
Compact
Standard
Extended
End-to-end
Ferritin
BSA
Lysozyme
RNase A
SARS-CoV-2 N
VEGF
Concanavalin A
MBP
Myoglobin
Pepsin
Streptavidin
TNF-α
Trypsin
Carbonic anhydrase II
Chymotrypsinogen
Cytochrome c
GFP
Papain
Subtilisin
Thrombin
Angiopoietin-2
FGF2
IFN-gamma
IL-17A
Influenza H3N2 HA
PDGF-BB
navigation-only (18)
both-pass (30)
static-only (12)
both-fail (18)
End-to-end (navigation):
all 3 profiles delivered (7)
≥1 profile undelivered (19)